

Вариант 23

1. Подготовьте рабочее окружение в соответствии с типом вашей операционной системы

- Установите Docker
- Выполните базовую настройку

```
C:\home\ntfray> docker --version
Docker version 27.5.1+dfsg4, build cab968b3
```

2. Изучите простейшие консольные команды и возможности Docker Desktop (см. лекцию), создать собственный контейнер docker/getting-started, открыть в браузере и изучить tutorial

```
C:\home\ntfray> docker run -dp 80:80 docker/getting-started
47225e4bd6ec05ccabf44a60e08e7741ce4c10a7be03ce904e2b97a7fa7949b3
C:\home\ntfray> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
47225e4bd6ec   docker/getting-started   "/docker-entrypoint..."   About a minute ago   Up About a minute   0.0.0.0:80->80/tcp, :::80->80/tcp   epic_blackburn
C:\home\ntfray>
```

Getting Started

The command you just ran

Congratulations! You have started the container for this tutorial! Let's first explain the command that you just ran. In case you forgot, here's the command:

```
docker run -d -p 80:80 docker/getting-started
```

Table of contents

- The command you just ran
- The Docker Dashboard
- What is a container?
- What is a container image?

```
C:\home\ntfray> docker info
Client:
Version:      27.5.1+dfsg4
Context:      default
Debug Mode:  false
Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  0.19.3+ds1
    Path:     /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  2.32.4-3
    Path:     /usr/libexec/docker/cli-plugins/docker-compose

Server:
Containers: 1
  Running: 1
  Paused: 0
  Stopped: 0
Images: 2
Server Version: 27.5.1+dfsg4
Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: false
  Native Overlay Diff: true
  userxattr: false
Logging Driver: json-file
Cgroup Driver: systemd
Cgroup Version: 2
Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
Swarm: inactive
Runtimes: io.containerd.runc.v2 runc
Default Runtime: runc
Init Binary: docker-init
containerd version: 1.7.24~ds1-10
runc version: 1.3.3+ds1-2
init version:
Security Options:
  apparmor
  seccomp
   Profile: builtin
  cgroupns
Kernel Version: 6.18.9+kali-amd64
Operating System: Kali GNU/Linux Rolling
OSType: linux
Architecture: x86_64
CPUs: 16
Total Memory: 15.03GiB
Name: myray
ID: 6dbbe81a-febd-4afc-ad7b-21a32f2b62dd
Docker Root Dir: /var/lib/docker
Debug Mode: false
Experimental: false
Insecure Registries:
  127.0.0.0/8
Live Restore Enabled: false
```

```
C:\home\ntfray> docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
hello-world          latest              1b44b5a3e06a       6 months ago       10.1kB
docker/getting-started latest             3e4394f6b72f       3 years ago        47MB

C:\home\ntfray> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED            STATUS             PORTS                               NAMES
47225e4bd6ec   docker/getting-started              "/docker-entrypoint..." 4 minutes ago      Up 4 minutes      0.0.0.0:80→80/tcp, :::80→80/tcp    epic_blackburn
C:\home\ntfray> docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED            STATUS             PORTS                               NAMES
47225e4bd6ec   docker/getting-started              "/docker-entrypoint..." 4 minutes ago      Up 4 minutes      0.0.0.0:80→80/tcp, :::80→80/tcp    epic_blackburn
C:\home\ntfray> docker ps -s
CONTAINER ID   IMAGE                                COMMAND                  CREATED            STATUS             PORTS                               NAMES             SIZE
47225e4bd6ec   docker/getting-started              "/docker-entrypoint..." 4 minutes ago      Up 4 minutes      0.0.0.0:80→80/tcp, :::80→80/tcp    epic_blackburn    1.09kB (virtual 47MB)
```

```
C:\home\ntfray> docker logs 47225e4bd6ec
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/02/14 09:16:18 [notice] 1#1: using the "epoll" event method
2026/02/14 09:16:18 [notice] 1#1: nginx/1.23.3
2026/02/14 09:16:18 [notice] 1#1: built by gcc 12.2.1 20220924 (Alpine 12.2.1_git20220924-r4)
2026/02/14 09:16:18 [notice] 1#1: OS: Linux 6.18.9-kali-amd64
2026/02/14 09:16:18 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 2147483584:2147483584
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2026/02/14 09:16:18 [notice] 1#1: start worker process 45
172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /tutorial/ HTTP/1.1" 304 0 "http://localhost/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /assets/stylesheets/application.adb8469c.css HTTP/1.1" 200 76332 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /assets/stylesheets/application-palette.a8b3c06d.css HTTP/1.1" 200 38773 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /assets/javascripts/modernizr.86422ebf.js HTTP/1.1" 200 7296 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /assets/fonts/material-icons.css HTTP/1.1" 200 873 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
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172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /assets/javascripts/application.c33a9706.js HTTP/1.1" 200 79589 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
```

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C:\home\ntfray> docker logs -f 47225e4bd6ec
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^CContext canceled
```

```
C:\home\ntfray> docker logs -t 47225e4bd6ec
2026-02-14T09:16:18.066935400Z /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
2026-02-14T09:16:18.066966478Z /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
2026-02-14T09:16:18.067468933Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
2026-02-14T09:16:18.073043700Z 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
2026-02-14T09:16:18.079244083Z 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
2026-02-14T09:16:18.07983137Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
2026-02-14T09:16:18.081480881Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
2026-02-14T09:16:18.082468208Z /docker-entrypoint.sh: Configuration complete; ready for start up
2026-02-14T09:16:18.088238343Z 2026/02/14 09:16:18 [notice] #1: using the "epoll" event method
2026-02-14T09:16:18.088314806Z 2026/02/14 09:16:18 [notice] #1: nginx/1.23.3
2026-02-14T09:16:18.088320547Z 2026/02/14 09:16:18 [notice] #1: built by gcc 12.2.1 20220924 (Alpine 12.2.1_git20220924-r4)
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2026-02-14T09:16:18.089876143Z 2026/02/14 09:16:18 [notice] #1: start worker process 34
2026-02-14T09:16:18.089982994Z 2026/02/14 09:16:18 [notice] #1: start worker process 35
2026-02-14T09:16:18.090123819Z 2026/02/14 09:16:18 [notice] #1: start worker process 36
2026-02-14T09:16:18.090245378Z 2026/02/14 09:16:18 [notice] #1: start worker process 37
2026-02-14T09:16:18.090383748Z 2026/02/14 09:16:18 [notice] #1: start worker process 38
2026-02-14T09:16:18.090511238Z 2026/02/14 09:16:18 [notice] #1: start worker process 39
2026-02-14T09:16:18.090650840Z 2026/02/14 09:16:18 [notice] #1: start worker process 40
2026-02-14T09:16:18.090786656Z 2026/02/14 09:16:18 [notice] #1: start worker process 41
2026-02-14T09:16:18.090954652Z 2026/02/14 09:16:18 [notice] #1: start worker process 42
2026-02-14T09:16:18.091123927Z 2026/02/14 09:16:18 [notice] #1: start worker process 43
2026-02-14T09:16:18.091239377Z 2026/02/14 09:16:18 [notice] #1: start worker process 44
2026-02-14T09:16:18.091381906Z 2026/02/14 09:16:18 [notice] #1: start worker process 45
2026-02-14T09:16:36.583076381Z 172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /tutorial/ HTTP/1.1" 304 0 "http://localhost/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.651763116Z 172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /assets/stylesheets/application.adb8469c.css HTTP/1.1" 200 76332 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.651966218Z 172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /assets/stylesheets/application-palette.a8b3c06d.css HTTP/1.1" 200 38773 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.654249520Z 172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /assets/javascripts/modernizr.86422ebf.js HTTP/1.1" 200 7296 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.658273271Z 172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /assets/fonts/material-icons.css HTTP/1.1" 200 873 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.658356127Z 172.17.0.1 - - [14/Feb/2026:09:16:36 +0000] "GET /css/styles.css HTTP/1.1" 200 892 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
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2026-02-14T09:16:36.658919127Z 172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/stylesheets/application.adb8469c.css HTTP/1.1" 200 76332 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.659178439Z 172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/javascripts/modernizr.86422ebf.js HTTP/1.1" 200 7296 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.659349520Z 172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/stylesheets/application-palette.a8b3c06d.css HTTP/1.1" 200 38773 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.659520520Z 172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/fonts/material-icons.css HTTP/1.1" 200 873 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.659698422Z 172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /css/styles.css HTTP/1.1" 200 892 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.659873357Z 172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/fonts/font-awesome.css HTTP/1.1" 200 30721 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.659998422Z 172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/javascripts/application.c33a9706.js HTTP/1.1" 200 79589 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.660173357Z 172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/javascripts/application.c33a9706.js HTTP/1.1" 200 79589 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
2026-02-14T09:16:36.660348422Z 172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/javascripts/application.c33a9706.js HTTP/1.1" 200 79589 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
```

```
C:\home\ntfray> docker logs --tail 5 47225e4bd6ec
172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/stylesheets/application-palette.a8b3c06d.css HTTP/1.1" 200 38773 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/fonts/material-icons.css HTTP/1.1" 200 873 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /css/styles.css HTTP/1.1" 200 892 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/fonts/font-awesome.css HTTP/1.1" 200 30721 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
172.17.0.1 - - [14/Feb/2026:09:22:57 +0000] "GET /assets/javascripts/application.c33a9706.js HTTP/1.1" 200 79589 "http://localhost/tutorial/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
C:\home\ntfray>
```

```
C:\home\ntfray> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                NAMES
47225e4bd6ec   docker/getting-started   "/docker-entrypoint..."   14 minutes ago   Up 14 minutes   0.0.0.0:80→80/tcp, :::80→80/tcp   epic_blackburn
C:\home\ntfray> docker pause 47225e4bd6ec
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                NAMES
47225e4bd6ec   docker/getting-started   "/docker-entrypoint..."   15 minutes ago   Up 15 minutes (Paused)   0.0.0.0:80→80/tcp, :::80→80/tcp   epic_blackburn
C:\home\ntfray> docker unpause 47225e4bd6ec
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                NAMES
47225e4bd6ec   docker/getting-started   "/docker-entrypoint..."   18 minutes ago   Up 17 minutes   0.0.0.0:80→80/tcp, :::80→80/tcp   epic_blackburn
```

```
C:\home\ntfray> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                NAMES
47225e4bd6ec   docker/getting-started   "/docker-entrypoint..."   19 minutes ago   Up 48 seconds   0.0.0.0:80→80/tcp, :::80→80/tcp   epic_blackburn
C:\home\ntfray> docker stop 47225e4bd6ec
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                NAMES
47225e4bd6ec   docker/getting-started   "/docker-entrypoint..."   19 minutes ago   Exited (0) 2 seconds ago   0.0.0.0:80→80/tcp, :::80→80/tcp   epic_blackburn
C:\home\ntfray> docker start 47225e4bd6ec
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                NAMES
47225e4bd6ec   docker/getting-started   "/docker-entrypoint..."   19 minutes ago   Up 3 seconds   0.0.0.0:80→80/tcp, :::80→80/tcp   epic_blackburn
C:\home\ntfray> docker restart 47225e4bd6ec
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                NAMES
47225e4bd6ec   docker/getting-started   "/docker-entrypoint..."   20 minutes ago   Up 6 seconds   0.0.0.0:80→80/tcp, :::80→80/tcp   epic_blackburn
C:\home\ntfray>
```

```
C:\home\ntfray> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                   NAMES
47225e4bd6ec   docker/getting-started   "/docker-entrypoint. ..."  23 minutes ago Up 3 seconds   0.0.0.0:80→80/tcp, :::80→80/tcp   epic_blackburn
C:\home\ntfray> docker stop 47225e4bd6ec
47225e4bd6ec
C:\home\ntfray> docker rm 47225e4bd6ec
47225e4bd6ec
C:\home\ntfray> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                   NAMES
C:\home\ntfray> docker images
REPOSITORY      TAG         IMAGE ID      CREATED        SIZE
docker/getting-started   latest     3e4394f6b72f   3 years ago    47MB
C:\home\ntfray> docker rmi 3e4394f6b72f
Untagged: docker/getting-started:latest
Untagged: docker/getting-started@sha256:d79336f4812b6547a53e735480dde67f8f7071b414fbd9297609ffb989abc1
Deleted: sha256:3e4394f6b72f7cfe4a2217067a77ff84d5d828afa0623867d68fce4f9d862b6c
Deleted: sha256:cdc6440a071be2985ce94c7e2e0c2df763b58a2ced4ecd944fcd9b13e7a2aa4
Deleted: sha256:041ac26cd02fa81c8fd73c616bdeee180de3fd68a649ed1c0339a84cdf7a723
Deleted: sha256:376baf7ada4b52ef4c110a023fe7185c4c2c090fa24a5cb4746066333ce3bc46
Deleted: sha256:d254c9b1e23bad05f5cde233b5e91153a0540fa9a797a580d8a360ad12bf63a9
Deleted: sha256:dd5c79fa9b6829fd08ff9943fcd66bebb3a3e04246ba394d57c28827fed95af0
Deleted: sha256:8d812a075abf60a83013c37f49058c220c9cdf390266952126e7e60041b305dc
Deleted: sha256:ff1787ee3dcae843dc7dd1933c49350fc84451cf19ed74f4ea72426e17ee7cd1
Deleted: sha256:85ebd294be1553b207ba9120676f4fd140842348ddf1bb5f7602c7a8401f0a13
Deleted: sha256:ded7a220bb058e28ee3254fbba04ca90b679070424424761a53a043b93b612bf
C:\home\ntfray> docker images
REPOSITORY      TAG         IMAGE ID      CREATED        SIZE
C:\home\ntfray>
```

```
C:\home\ntfray> docker images
REPOSITORY      TAG         IMAGE ID      CREATED        SIZE
C:\home\ntfray> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                   NAMES
C:\home\ntfray> docker pull docker/getting-started
Using default tag: latest
latest: Pulling from docker/getting-started
c158987b0551: Pull complete
1e35f6679fab: Pull complete
cb9626c74200: Pull complete
b6334b6ace34: Pull complete
f1d1c9928c82: Pull complete
9b6f639ec6ea: Pull complete
ee68d3549ec8: Pull complete
33e0cbbb4673: Pull complete
4f7e34c2de10: Pull complete
Digest: sha256:d79336f4812b6547a53e735480dde67f8f7071b414fbd9297609ffb989abc1
Status: Downloaded newer image for docker/getting-started:latest
docker.io/docker/getting-started:latest
C:\home\ntfray> docker images
REPOSITORY      TAG         IMAGE ID      CREATED        SIZE
docker/getting-started   latest     3e4394f6b72f   3 years ago    47MB
C:\home\ntfray> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                   NAMES
C:\home\ntfray> docker create -p 80:80 3e4394f6b72f
8b338af6bce4955d024ee1a25bc147e02ac034f7f54f8c7cf53063af66c5d774
C:\home\ntfray> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                   NAMES
8b338af6bce4   3e4394f6b72f   "/docker-entrypoint. ..."  4 seconds ago Created                xenodochial_cor1
C:\home\ntfray> docker start 8b338af6bce4
8b338af6bce4
C:\home\ntfray> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                   NAMES
8b338af6bce4   3e4394f6b72f   "/docker-entrypoint. ..."  50 seconds ago Up 7 seconds   0.0.0.0:80→80/tcp, :::80→80/tcp   xenodochial_cor1
C:\home\ntfray>
```

```
C:\home\ntfray> docker rm 4438ba90d08b
4438ba90d08b
C:\home\ntfray> docker run -dp 80:80 --name tutor docker/getting-started
0e6c55f1a387ee502c821be3943349ec05b308ef98bfc41519dd2c27c5d8cf61
C:\home\ntfray> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                   NAMES
0e6c55f1a387   docker/getting-started   "/docker-entrypoint. ..."  4 seconds ago Up 3 seconds   0.0.0.0:80→80/tcp, :::80→80/tcp   tutor
C:\home\ntfray>
```

```
C:\home\ntfray> docker run -it --name testPython python
Unable to find image 'python:latest' locally
latest: Pulling from library/python
ef235bf1a09a: Pull complete
954d6059ca7b: Pull complete
b5e2021c4c8b: Pull complete
128c71264009: Pull complete
d8871274053b: Pull complete
4f69a3eb488f: Pull complete
c59471c320a2: Pull complete
Digest: sha256:151ab3571dad616bb031052e86411e2165295c7f67ef27206852203e854bcd12
Status: Downloaded newer image for python:latest
Python 3.14.3 (main, Feb  4 2026, 20:08:31) [GCC 14.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> 1 + 2
3
>>> print("Hello, World!")
Hello, World!
>>> 2 + 2 * 2
6
```

```
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
docker/getting-started latest          3e4394f6b72f   3 years ago    47MB
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> docker tag 3e4394f6b72f tutotTag
Error parsing reference: "tutotTag" is not a valid repository/tag: invalid reference format: repository name (library/tutotTag) must be lowercase
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> docker tag 3e4394f6b72f tutortag
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
docker/getting-started latest          3e4394f6b72f   3 years ago    47MB
tutortag             latest          3e4394f6b72f   3 years ago    47MB
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> docker rmi docker/getting-started
Untagged: docker/getting-started:latest
Untagged: docker/getting-started@sha256:d79336f4812b6547a53e735480dde67f8f7071b414fbd9297609ffb989abc1
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
tutortag             latest          3e4394f6b72f   3 years ago    47MB
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> docker rmi tutortag
Error response from daemon: conflict: unable to remove repository reference "tutortag" (must force) - container 4feafac2794e is using its reference
d image 3e4394f6b72f
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> █
```

3. Создайте docker image, который запускает скрипт с использованием функций из https://github.com/smartikaorg/geometric_lib.

1. Данные необходимые для работы скрипта передайте любым удобным способом (например: конфиг файл через docker volume, переменные окружения, перенаправление ввода). Изучите простейшие консольные команды для работы с docker(см. лекцию). Зарегистрируйтесь на DockerHub и выберите необходимые для проекта образы
2. Создать Dockerfile для реализации сборки собственных Docker образов
3. Использовать его для создания контейнера. Протестировать использование контейнера

```
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> ls
Dockerfile  LR2Test.txt  script.py  'ИГМ-2 2024.pptx'  'Лабораторная работа 2.docx'
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> docker build -t scriptpy -f Dockerfile .
[+] Building 1.4s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 157B
=> [internal] load metadata for docker.io/library/python:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/python:latest@sha256:151ab3571dad616bb031052e86411e2165295c7f67ef27206852203e854bcd12
=> [internal] load build context
=> => transferring context: 3.65kB
=> CACHED [2/4] WORKDIR /app
=> CACHED [3/4] COPY LR1/geometric_lib /app/geometric_lib
=> [4/4] COPY LR2/script.py /app/
=> exporting to image
=> => exporting layers
=> => writing image sha256:4eb97018778eea9efe865c643c363aef84ad1a30cc967f1ce25bd3d51952d143
=> => naming to docker.io/library/scriptpy
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> docker run scriptpy

=====

Circle (radius=5):
Area: 78.54
Perimeter: 31.42

Square (side=4):
Area: 16
Perimeter: 16
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> docker run -e RADIUS=120 -e SIDE=200 scriptpy

=====

Circle (radius=120):
Area: 45238.93
Perimeter: 753.98

Square (side=200):
Area: 40000
Perimeter: 800
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> █
```

```
import sys
import os
sys.path.append('/app/geometric_lib')

from circle import area as circle_area, perimeter as circle_perimeter
from square import area as square_area, perimeter as square_perimeter

radius = int(os.getenv('RADIUS', '5'))
side = int(os.getenv('SIDE', '4'))

print("\n===== ")
print(f"\nCircle (radius={radius}):")
print(f"Area: {circle_area(radius):.2f}")
print(f"Perimeter: {circle_perimeter(radius):.2f}")
print(f"\nSquare (side={side}):")
print(f"Area: {square_area(side):.2f}")
print(f"Perimeter: {square_perimeter(side):.2f}")
```

```
FROM python
WORKDIR /app
COPY LR1/geometric_lib /app/geometric_lib
COPY LR2/script.py /app/
CMD ["python", "script.py"]
```

4. Скачать любой доступный проект с GitHub с произвольным стеком технологий (пример – см. индивидуальное задание) или использовать свой, ранее разработанный. Для моего 23 варианта MEAN stack: MongoDB, Express.js, AngularJS, Node.js. Создать для него необходимый контейнер, используя Docker Compose для управления многоконтейнерными приложениями. Запустить проект в контейнере. (Примеры Images: https://hub.docker.com/_/phpmyadmin, https://hub.docker.com/_/mysql, https://hub.docker.com/_/postgres).

```
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> ls
app  config.json  controllers  LICENSE  package.json  README.md  server.js  services  views
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> vim Dockerfile
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> ls
app  config.json  controllers  Dockerfile  LICENSE  package.json  README.md  server.js  services  views
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> vim docker-compose.yml
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> ls
app  config.json  controllers  docker-compose.yml  Dockerfile  LICENSE  package.json  README.md  server.js  services  views
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> docker-compose up --build
WARN[0000] /home/ntfray/IGI_Labs/453504_Romanov_23/IGI/LR2/mean-stack-registration-login-example/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 14/14
  mongo  Pulled 31.4s
  ✓ 924737f7ef455 Pull complete 13.3s
  ✓ fb52bde70123 Pull complete 13.3s
  ✓ 64788f86be3f Pull complete 13.3s
  ✓ 33fd5f2e001 Pull complete 13.3s
  ✓ 570e5665608 Pull complete 13.3s
  ✓ f518a872ab12 Pull complete 13.6s
  ✓ c9bdae151f64 Pull complete 13.7s
  ✓ b2c58d5f563 Pull complete 13.7s
  ✓ 2928038a6053 Pull complete 13.7s
  ✓ 29a16c1b79ab Pull complete 13.8s
  ✓ efec0f86077c Pull complete 26.6s
  ✓ 261a04726d31 Pull complete 26.6s
  ✓ 7c4062426e72 Pull complete 26.6s
[+] Building 66.2s (11/11) FINISHED docker:default
=> [app internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 150B 0.0s
=> [app internal] load metadata for docker.io/library/node:8 1.8s
=> [app internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [app 1/5] FROM docker.io/library/node:8sha256:a681bf74805b80d03eb21a6c0ef168a976108a287a74167ab593fc953aac34df 56.3s
=> => resolve docker.io/library/node:8sha256:a681bf74805b80d03eb21a6c0ef168a976108a287a74167ab593fc953aac34df 0.0s
=> => sha256:a681bf74805b80d03eb21a6c0ef168a976108a287a74167ab593fc953aac34df 1.42kB / 1.42kB 0.0s
=> => sha256:146b6da886182fde06bf747470b1c89814bc8a1c96fd1aef6107171959f6e 45.38MB / 45.38MB 16.4s
=> => sha256:db0efb8e06081b5b0bb7c406426982c4202839687c1ac3941b364527e2249 4.34MB / 4.34MB 2.1s
=> => sha256:ee25271e00572cb09d095fb35880a284eac570c1745264db6c7f1d08c7f9f7 2.21kB / 2.21kB 0.0s
=> => sha256:beeadf3757f41ffdbefb40bbe14f9f5b9ce33c1a88bb5c0b4d6e9a67509d 7.76kB / 7.76kB 0.0s
=> => sha256:9935d0c62ace92b388be202275e222007d6cac10b9c1f2c1ea63af38c09ea7ab 10.80MB / 10.80MB 3.9s
=> => sha256:c705a4cfd310b96bf3d7928428e65f0d3f5bad0c0bda143aae1d89418468 50.07MB / 50.07MB 19.7s
=> => sha256:c87b722db6f4d4297c21f07162d21fa6c6f51d03360f75e24cfa3bcae57d 214.84MB / 214.84MB 42.2s
=> => extracting sha256:146b6da886182fde06bf747470b1c89814bc8a1c96fd1aef6107171959f6e 3.2s
=> => sha256:db0efb8e06081b5b0bb7c406426982c4202839687c1ac3941b364527e2249 4.16kB / 4.16kB 16.8s
=> => sha256:db0efb8e06081b5b0bb7c406426982c4202839687c1ac3941b364527e2249 19.40MB / 19.40MB 22.7s
=> => sha256:1c151cd1b3eabc8d84742529489b3c8c344181b4cbdcfcddbf4a7c7d0239ec 1.40MB / 1.40MB 20.5s
=> => extracting sha256:9935d0c62ace92b388be202275e222007d6cac10b9c1f2c1ea63af38c09ea7ab 0.6s
=> => sha256:fb0993995f407f3681fc8399ae67c04bba8e15c7f754b1af8b37f50b35fa11e 292B / 292B 20.8s
=> => extracting sha256:db0efb8e06081b5b0bb7c406426982c4202839687c1ac3941b364527e2249 3.6s
=> => extracting sha256:e705a4cfd310b96bf3d7928428e65f0d3f5bad0c0bda143aae1d89418468 0.2s
=> => extracting sha256:c87b722db6f4d4297c21f07162d21fa6c6f51d03360f75e24cfa3bcae57d 11.6s
=> => extracting sha256:db0efb8e06081b5b0bb7c406426982c4202839687c1ac3941b364527e2249 1.9s
=> => extracting sha256:1c151cd1b3eabc8d84742529489b3c8c344181b4cbdcfcddbf4a7c7d0239ec 0.1s
=> => extracting sha256:fb0993995f407f3681fc8399ae67c04bba8e15c7f754b1af8b37f50b35fa11e 0.0s
=> [app internal] load build context 0.0s
=> => transferring context: 86.81kB 0.0s
=> [app 2/5] WORKDIR /app 0.7s
=> [app 3/5] COPY package.json /app/ 0.0s
=> [app 4/5] RUN npm install --silent 6.8s
=> [app 5/5] COPY . 0.0s
=> [app] exporting to image 0.4s
=> => exporting layers 0.4s
=> => writing image sha256:7fc4276a9157922ce67ccc290cba3ffb10cf26bf44d78366bd1026938caf49 0.0s
=> => naming to docker.io/library/mean-stack-registration-login-example-app 0.0s
```

```
=> => naming to docker.io/library/mean-stack-registration-login-example-app 0.0s
=> [app] resolving provenance for metadata file 0.0s
[+] Running 4/4
  app  Built 0.0s
  ✓ Network mean-stack-registration-login-example-default Created 0.2s
  ✓ Container mean-stack-registration-login-example-mongo-1 Created 0.0s
  ✓ Container mean-stack-registration-login-example-app-1 Created 0.0s
Attaching to app-1, mongo-1
mongo-1 | 2026-02-15T13:14:20.946+0000 I CONTROL [initandlisten] MongoDB starting : pid=1 port=27017 dbpath=/data/db 64-bit host=7bf80755dad3
mongo-1 | 2026-02-15T13:14:20.946+0000 I CONTROL [initandlisten] db version v3.6.23
mongo-1 | 2026-02-15T13:14:20.946+0000 I CONTROL [initandlisten] git version: d352e6a4764659e0d0350ce77279de3c1f243e5c
mongo-1 | 2026-02-15T13:14:20.946+0000 I CONTROL [initandlisten] OpenSSL version: OpenSSL 1.0.2g 1 Mar 2016
mongo-1 | 2026-02-15T13:14:20.946+0000 I CONTROL [initandlisten] allocator: tcmalloc
mongo-1 | 2026-02-15T13:14:20.946+0000 I CONTROL [initandlisten] modules: none
mongo-1 | 2026-02-15T13:14:20.946+0000 I CONTROL [initandlisten] build environment:
mongo-1 | 2026-02-15T13:14:20.946+0000 I CONTROL [initandlisten] distmod: ubuntu1604
mongo-1 | 2026-02-15T13:14:20.946+0000 I CONTROL [initandlisten] distarch: x86_64
mongo-1 | 2026-02-15T13:14:20.946+0000 I CONTROL [initandlisten] target_arch: x86_64
mongo-1 | 2026-02-15T13:14:20.946+0000 I CONTROL [initandlisten] options: { net: { bindIpAll: true } }
mongo-1 | 2026-02-15T13:14:20.948+0000 I STORAGE [initandlisten]
mongo-1 | 2026-02-15T13:14:20.949+0000 I STORAGE [initandlisten] ** WARNING: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine
mongo-1 | 2026-02-15T13:14:20.949+0000 I STORAGE [initandlisten] See http://dochub.mongodb.org/core/prodnotes-filesystem
mongo-1 | 2026-02-15T13:14:20.949+0000 I STORAGE [initandlisten] wiredtiger-open config: create,cache_size=7183M,cache_overflow=(file_max=0M),session_max=20000,eviction=(threads_min=4,threads_max=4),config_base=false,statistics=(fast),compatibility=(release='3.0',require_max='3.0'),log=(enabled=true,archive=true,path=journal,compressor=snappy),file_manager=(close_idle_time=100000),statistics_log=(wait=0),verbose=(recovery_progress),
mongo-1 | 2026-02-15T13:14:21.162+0000 I STORAGE [initandlisten] WiredTiger message [1771161261:162230][1:0x7fd3bcde0a40], txn-recover: Set global recovery timestamp: 0
mongo-1 | 2026-02-15T13:14:21.183+0000 I CONTROL [initandlisten]
mongo-1 | 2026-02-15T13:14:21.183+0000 I CONTROL [initandlisten] ** WARNING: Access control is not enabled for the database.
mongo-1 | 2026-02-15T13:14:21.183+0000 I CONTROL [initandlisten] ** Read and write access to data and configuration is unrestricted.
mongo-1 | 2026-02-15T13:14:21.184+0000 I CONTROL [initandlisten]
mongo-1 | 2026-02-15T13:14:21.184+0000 I CONTROL [initandlisten] ** WARNING: /sys/kernel/mm/transparent_hugepage/enabled is 'always'.
mongo-1 | 2026-02-15T13:14:21.184+0000 I CONTROL [initandlisten] ** We suggest setting it to 'never'.
mongo-1 | 2026-02-15T13:14:21.184+0000 I STORAGE [initandlisten] createCollection: admin.system.version with provided UUID: b88bba7e-afe9-4ef9-81d5-ca6f472fdc26
mongo-1 | 2026-02-15T13:14:21.192+0000 I COMMAND [initandlisten] setting featureCompatibilityVersion to 3.6
mongo-1 | 2026-02-15T13:14:21.193+0000 I STORAGE [initandlisten] createCollection: local.startup_log with generated UUID: 1ee075ac-302a-4369-a80c-esb68119013a
mongo-1 | 2026-02-15T13:14:21.202+0000 I FTDC [initandlisten] Initializing full-time diagnostic data capture with directory '/data/db/diagnostic.data'
mongo-1 | 2026-02-15T13:14:21.202+0000 I NETWORK [initandlisten] listening via socket bound to 0.0.0.0
mongo-1 | 2026-02-15T13:14:21.202+0000 I STORAGE [LogicalSessionCacheRefresh] createCollection: config.system.sessions with generated UUID: e0ec0144-dc24-4b23-b2ed-49db1d2c64c4
mongo-1 | 2026-02-15T13:14:21.202+0000 I NETWORK [initandlisten] listening via socket bound to /tmp/mongod-27017.sock
mongo-1 | 2026-02-15T13:14:21.202+0000 I NETWORK [initandlisten] waiting for connections on port 27017
mongo-1 | 2026-02-15T13:14:21.214+0000 I INDEX [LogicalSessionCacheRefresh] build index on: config.system.sessions properties: { v: 2, key: { lastUse: 1 }, name: "lsid TTLIndex", ns: "config.system.sessions", expireAfterSeconds: 1800 }
mongo-1 | 2026-02-15T13:14:21.214+0000 I INDEX [LogicalSessionCacheRefresh] building index using bulk method; build may temporarily use up to 500 megabytes of RAM
mongo-1 | 2026-02-15T13:14:21.215+0000 I INDEX [LogicalSessionCacheRefresh] build index done. scanned 0 total records. 0 secs
app-1 | Server listening at http://:::3000
```

Enable Watch

http://localhost:3000/login

METANIT

MDN

Yandex Music

Monkeytype

YouTube

Интегралы

DeepSeek

Grok

СЗО

x1

x3

x4

Mail

Login

Username

Password

Login

Register

MEAN Stack User Registration and Login Example

JasonWatmore.com

http://localhost:3000/register

METANIT

MDN

Yandex Music

Monkeytype

YouTube

Интегралы

DeepSeek

Grok

СЗО

x1

x3

x4

Mail

Register

First name

Nikolay

Last name

Romanov

Username

NikolayRom

Password

Register

Cancel

MEAN Stack User Registration and Login Example

JasonWatmore.com

http://localhost:3000/login

METANIT

MDN

Yandex Music

Monkeytype

YouTube

Интегралы

DeepSeek

Grok

СЗО

x1

x3

x4

Mail

Login

Registration successful

Username

Password

Login

Register

MEAN Stack User Registration and Login Example

JasonWatmore.com

←

→

↺

🏠

http://localhost:3000/app/#/

☆

METANIT

MDN

Yandex Music

Monkeytype

YouTube

Интергалы

DeepSeek

Grok

СЭО

x1

x3

x4

Mail

Home

Account

Logout

Hi Nikolay!!

←

→

↺

🏠

http://localhost:3000/app/#/account

☆

METANIT

MDN

Yandex Music

Monkeytype

YouTube

Интергалы

DeepSeek

Grok

СЭО

x1

x3

x4

Mail

Home

Account

Logout

My Account

First name

Nikolay

Last name

Romanov

Username

NikolayRom

Password

Save

Delete

```
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> ls
Dockerfile  LR2Test.txt  mean-stack-registration-login-example  script.py  'ИГИ-2 2024.pptx'  'Лабораторная работа 2.docx'
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2> cd mean-stack-registration-login-example
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> ls
app  config.json  controllers  data  docker-compose.yml  Dockerfile  LICENSE  package.json  README.md  server.js  services  views
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> cd data
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example\data> ls
collection-0-3947945325138800506.wt  diagnostic.data  index-6-3947945325138800506.wt  mongod.lock  WiredTigerLAS.wt
collection-2-3947945325138800506.wt  index-1-3947945325138800506.wt  index-8-3947945325138800506.wt  sizeStorer.wt  WiredTiger.lock
collection-4-3947945325138800506.wt  index-3-3947945325138800506.wt  journal  storage.bson  WiredTiger.turtle
collection-7-3947945325138800506.wt  index-5-3947945325138800506.wt  _mdb_catalog.wt  WiredTiger  WiredTiger.wt

app-1 | Server listening at http://:::3000
mongo-1 | 2026-02-15T13:19:01.058+0000 I NETWORK [listener] connection accepted from 172.18.0.3:52728 #1 (1 connection now open)
mongo-1 | 2026-02-15T13:19:01.063+0000 I NETWORK [conn1] received client metadata from 172.18.0.3:52728 conn1: { driver: { name: "nodejs", version: "2.2.36" }, os: { typ
e: "linux", name: "linux", architecture: "x64", version: "6.18.9+kali-amd64" }, platform: "Node.js v8.17.0, LE, mongodb-core: 2.1.20" }
mongo-1 | 2026-02-15T13:19:01.223+0000 I STORAGE [conn1] createCollection: mean-stack-registration-login-example.users with generated UUID: 422dia32-d106-4f01-b5e1-e9e79
ebeb769
[]

mongo-1 | 2026-02-15T13:19:01.223+0000 I STORAGE [conn1] createCollection: mean-stack-registration-login-example.users with generated UUID: 422dia32-d106-4f01-b5e1-e9e79
ebeb769
Gracefully stopping... (press Ctrl+C again to force)
[*] Stopping 2/2
✓ Container mean-stack-registration-login-example-app-1 Stopped 10.2s
✓ Container mean-stack-registration-login-example-mongo-1 Stopped 0.2s
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example>

CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
8a400453da4b mean-stack-registration-login-example-app "docker-entrypoint.s..." 17 minutes ago Exited (137) About a minute ago
7bf60755dad3 mongo:3.6 "docker-entrypoint.s..." 17 minutes ago Exited (0) About a minute ago mean-stack-registration-login-example-mongo-1

C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> docker-compose up -d
WARN[0000] /home/ntfray/IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[*] Running 2/2
✓ Container mean-stack-registration-login-example-mongo-1 Started 0.3s
✓ Container mean-stack-registration-login-example-app-1 Started 0.6s
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
8a400453da4b mean-stack-registration-login-example-app "docker-entrypoint.s..." 19 minutes ago Up 28 seconds 0.0.0.0:3000->3000/tcp, :::3000->3000/tcp mean-stack-registration-login-example-app-1
7bf60755dad3 mongo:3.6 "docker-entrypoint.s..." 19 minutes ago Up 29 seconds 27017/tcp mean-stack-registration-login-example-mongo-1
4feaf6c2794e 3a3394f6b72f "/docker-entrypoint..." 28 hours ago Up 2 hours 0.0.0.0:80->80/tcp, :::80->80/tcp tutor
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example>
```

```

Version: '3'

services:

  app:
    build: .
    ports:
      - "3000:3000"
    links:
      - mongo
    depends_on:
      - mongo

  mongo:
    image: mongo:3.6
    volumes:
      - ./data:/data/db

```

```

FROM node:8
WORKDIR /app
COPY package.json /app/
RUN npm install --silent
COPY . .
EXPOSE 3000
CMD ["node", "server.js"]

```

```

C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> docker-compose down
[+] Running 2/2
✓ Container mean-stack-registration-login-example-app-1   Removed
✓ Container mean-stack-registration-login-example-mongo-1 Removed
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example>

```

```

C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> docker-compose up -d --build
[+] Building 2.5s (11/11) FINISHED
=> [app internal] load build definition from Dockerfile
=> => transferring dockerfile: 158B
=> [app internal] load metadata for docker.io/library/node:8
=> [app internal] load .dockerignore
=> => transferring context: 61B
=> [app 1/5] FROM docker.io/library/node:8@sha256:a681bf74805b80d03eb21a6c0ef168a976108a287a74167ab593fc953aac34df
=> [app internal] load build context
=> => transferring context: 26.98kB
=> CACHED [app 2/5] WORKDIR /app
=> CACHED [app 3/5] COPY package.json /app/
=> CACHED [app 4/5] RUN npm install --silent
=> [app 5/5] COPY . .
=> [app] exporting to image
=> => exporting layers
=> => writing image sha256:5a6149c336423a7fdca11e38ccf1e340a32cf95c17b0e02074db077acb733535
=> => naming to docker.io/library/mean-stack-registration-login-example-app
=> [app] resolving provenance for metadata file
[+] Running 4/4
✓ app                               Built
✓ Network mean-stack-registration-login-example_mean-network Created
✓ Container mean-stack-registration-login-example-mongo-1 Started
✓ Container mean-stack-registration-login-example-app-1 Started
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example>

```

5. Настроить сети и тома для обеспечения связи между контейнерами и сохранения данных (исходные данные, логин, пароль и т.д.)

```
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> docker network ls
NETWORK ID          NAME                                     DRIVER  SCOPE
a7812ddc1c8b        bridge                                bridge  local
0830714d6018        host                                 host    local
b3ac90e62daf        mean-stack-registration-login-example_default  bridge  local
9f2cbfd0cc8f        mean-stack-registration-login-example_mean-network  bridge  local
bbd18a792dbf        none                                 null    local
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> 
```

```
services:

  app:

    build: .

    ports:
      - "3000:3000"

    depends_on:
      - mongo

    networks:
      - mean-network

  mongo:

    image: mongo:3.6

    networks:
      - mean-network

    volumes:
      - ./data:/data/db

networks:

  mean-network:

    driver: bridge
```

6. Разместите результат в созданный репозиторий в DockerHub

```
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> docker login -u ntfray
Password:
WARNING! Your password will be stored unencrypted in /home/ntfray/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credential-stores

Login Succeeded
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
mean-stack-registration-login-example-app  latest             5a6149c33642       14 minutes ago     921MB
<none>              <none>             7fc4276a9157       About an hour ago   921MB
scriptpy            latest             4eb97018778e       25 hours ago        1.12GB
tutortag            latest             3e4394f6b72f       3 years ago         47MB
mongo               3.6                2f21415cb85f       4 years ago         453MB
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> docker tag mean-stack-registration-login-example-app:latest ntfray/igi-labs:mean-app
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
mean-stack-registration-login-example-app  latest             5a6149c33642       18 minutes ago     921MB
ntfray/igi-labs     mean-app            5a6149c33642       18 minutes ago     921MB
<none>              <none>             7fc4276a9157       About an hour ago   921MB
scriptpy            latest             4eb97018778e       25 hours ago        1.12GB
tutortag            latest             3e4394f6b72f       3 years ago         47MB
mongo               3.6                2f21415cb85f       4 years ago         453MB
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> docker push ntfray/igi-labs:mean-app
The push refers to repository [docker.io/ntfray/igi-labs]
5f696f73ae15: Pushed
c6f1856712cf: Pushed
8ce20ca9180c: Pushed
b44c7f25e553: Pushed
423451ed44f2: Mounted from library/node
b2aaf85d6633: Mounted from library/node
88601a85ce11: Mounted from library/node
42f9c2f9c08e: Mounted from library/node
99e8bd38fafc: Mounted from library/node
bee1e39d7c3a: Mounted from library/node
1f59a4b2e206: Mounted from library/node
0ca7f54856c0: Mounted from library/node
ebb9ae013834: Mounted from library/node
mean-app: digest: sha256:6f0b2c2088fafc0d181625cca01222ad5a88265a0c95ad212e1ed3440139f7ee size: 3047
C:\home\ntfray\IGI_Labs\453504_Romanov_23\IGI\LR2\mean-stack-registration-login-example> 
```

The screenshot shows the Docker Hub web interface for the repository `ntfray/igi-labs`. The main content area displays the repository's details, including its last push time and size. A table lists the tags, showing the `mean-app` tag with its OS, type, and push time. The sidebar on the left contains navigation links for repository management. On the right, there are sections for Docker commands and a Build Cloud advertisement.

7. Выполните следующие действия с целью изучить особенности сетевого взаимодействия:

- Получить информацию о всех сетях, работающих на текущем хосте и подробности о каждом типе сети
- Создать свою собственную сеть `bridge`, проверить, создана ли она, запустить Docker-контейнер в созданной сети, вывести о ней всю информацию (включая IP-адрес контейнера), отключить сеть от контейнера
- Создать еще одну сеть `bridge`, вывести о ней всю информацию, запустить в ней три контейнера, подключиться к любому из контейнеров и пропинговать два других из оболочки контейнера, убедиться, что между контейнерами происходит общение по IP-адресу
- Создать свою собственную сеть `overlay`, проверить, создана ли она, вывести о ней всю информацию
- Создать еще одну сеть `overlay`, проверить, создана ли она, вывести о ней всю информацию, удалить сеть

- Попробовать создать сеть host, сохранить результат в отчет.

```

→ LR2 git:(IGI-LR2) x docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
2dd2caad52b5        bridge             bridge             local
0830714d6018        host              host              local
b3ac90e62daf        mean-stack-registration-login-example_default    bridge             local
9f2cbfd0cc8f        mean-stack-registration-login-example_mean-network bridge             local
bbd18a792dbf        none              null              local
→ LR2 git:(IGI-LR2) x docker network inspect bridge host none
[
  {
    "Name": "bridge",
    "Id": "2dd2caad52b5571dbb534af339ec598639da3a2fd7f8cda49f3a3d5f57f29e32",
    "Created": "2026-02-15T23:12:52.862142441+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.17.0.0/16",
          "Gateway": "172.17.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {
      "com.docker.network.bridge.default_bridge": "true",
      "com.docker.network.bridge.enable_icc": "true",
      "com.docker.network.bridge.enable_ip_masquerade": "true",
      "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
      "com.docker.network.bridge.name": "docker0",
      "com.docker.network.driver.mtu": "1500"
    },
    "Labels": {}
  },
]

```

```

{
  "Name": "host",
  "Id": "0830714d6018eb7f307e1111d7a2032674a99c096c4978bac208d4a10ec92e42",
  "Created": "2026-02-13T14:29:59.216374425+03:00",
  "Scope": "local",
  "Driver": "host",
  "EnableIPv6": false,
  "IPAM": {
    "Driver": "default",
    "Options": null,
    "Config": null
  },
  "Internal": false,
  "Attachable": false,
  "Ingress": false,
  "ConfigFrom": {
    "Network": ""
  },
  "ConfigOnly": false,
  "Containers": {},
  "Options": {},
  "Labels": {}
},
{
  "Name": "none",
  "Id": "bbd18a792dbf6e27a2fff83daca65d83b409f675cc76d13bfd4e09aeb109f4d6",
  "Created": "2026-02-13T14:29:59.185494592+03:00",
  "Scope": "local",
  "Driver": "null",
  "EnableIPv6": false,
  "IPAM": {
    "Driver": "default",
    "Options": null,
    "Config": null
  },
  "Internal": false,
  "Attachable": false,
  "Ingress": false,
  "ConfigFrom": {
    "Network": ""
  },
  "ConfigOnly": false,
  "Containers": {},
  "Options": {},
  "Labels": {}
}
]
→ LR2 git:(IGI-LR2) x

```

```

→ LR2 git:(IGI-LR2) x docker network create -d bridge my-net-bridge
77241a4ca589033520a679a501c0b71a0e03a6ffd4331efb93dabd8f8cdad04e
→ LR2 git:(IGI-LR2) x docker network ls

```

NETWORK ID	NAME	DRIVER	SCOPE
2dd2caad52b5	bridge	bridge	local
0830714d6018	host	host	local
b3ac90e62daf	mean-stack-registration-login-example_default	bridge	local
9f2cbfd0cc8f	mean-stack-registration-login-example_mean-network	bridge	local
77241a4ca589	my-net-bridge	bridge	local
bbd18a792dbf	none	null	local

```

→ LR2 git:(IGI-LR2) x

```

```

→ LR2 git:(IGI-LR2) x docker run -dp 80:80 --name tutor-mynet --network my-net-bridge tutortag:latest
246826ebd69568366f3ce738e17ed496344dca32ecb6029fd2aea50fb68b7e
→ LR2 git:(IGI-LR2) x docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
246826ebd69    tutortag:latest  "/docker-entrypoint.…"  7 seconds ago  Up 7 seconds  0.0.0.0:80->80/tcp, :::80->80/tcp  tutor-mynet
→ LR2 git:(IGI-LR2) x docker network inspect my-net-bridge
[
  {
    "Name": "my-net-bridge",
    "Id": "77241a4ca589033520a679a501c0b71a0e03a6ffd4331efb93dab8f8cdaa04e",
    "Created": "2026-02-16T00:14:38.695687204+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "Gateway": "172.20.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "246826ebd69568366f3ce738e17ed496344dca32ecb6029fd2aea50fb68b7e": {
        "Name": "tutor-mynet",
        "EndpointID": "c6dca8c8c3b9bfcbb0a8d3c243092fe646e71dfbe3f5005ea7faa0c626b1da7",
        "MacAddress": "02:42:ac:14:00:02",
        "IPv4Address": "172.20.0.2/16",
        "IPv6Address": ""
      }
    },
    "Options": {},
    "Labels": {}
  }
]
→ LR2 git:(IGI-LR2) x

```

```

→ LR2 git:(IGI-LR2) x docker network disconnect my-net-bridge tutor-mynet
→ LR2 git:(IGI-LR2) x docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
246826ebd69    tutortag:latest  "/docker-entrypoint.…"  5 minutes ago  Up 5 minutes  0.0.0.0:80->80/tcp, :::80->80/tcp  tutor-mynet
→ LR2 git:(IGI-LR2) x docker network inspect my-net-bridge
[
  {
    "Name": "my-net-bridge",
    "Id": "77241a4ca589033520a679a501c0b71a0e03a6ffd4331efb93dab8f8cdaa04e",
    "Created": "2026-02-16T00:14:38.695687204+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "Gateway": "172.20.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]
→ LR2 git:(IGI-LR2) x

```

```
→ LR2 git:(IGI-LR2) x docker network create -d bridge my-net-ping
24511b00c9e54d365819fd5e6e86c491fabe865a1c596ae19313e22bf92853c9
→ LR2 git:(IGI-LR2) x docker network inspect my-net-ping
[
  {
    "Name": "my-net-ping",
    "Id": "24511b00c9e54d365819fd5e6e86c491fabe865a1c596ae19313e22bf92853c9",
    "Created": "2026-02-16T00:44:36.113399353+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.21.0.0/16",
          "Gateway": "172.21.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]
→ LR2 git:(IGI-LR2) x
```

```

→ LR2 git:(IGI-LR2) ✗ docker run -itd --name A1 --network my-net-ping alpine ash
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
589002ba0eae: Pull complete
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
6909b29992d5e8acdb0a5100606bd9e38431be784086b5a2dd0ac637c363a75c
→ LR2 git:(IGI-LR2) ✗ docker run -itd --name A2 --network my-net-ping alpine ash
ef9b53352bc238dba1ddc8796749c6eab5bdf80cc1f135b556d82bebcb768a92
→ LR2 git:(IGI-LR2) ✗ docker run -itd --name A3 --network my-net-ping alpine ash
ace50ccb5f5824a3388498b996774673228d493c64d72e1f8277da02126919bcd
→ LR2 git:(IGI-LR2) ✗ docker ps
CONTAINER ID   IMAGE      COMMAND      CREATED          STATUS          PORTS          NAMES
ace50ccb5f582  alpine    "ash"        3 seconds ago   Up 3 seconds           A3
ef9b53352bc2  alpine    "ash"        9 seconds ago   Up 9 seconds           A2
6909b29992d5  alpine    "ash"        18 seconds ago  Up 18 seconds          A1
→ LR2 git:(IGI-LR2) ✗ docker container attach A2
/ # ping -c 2 A1
PING A1 (172.21.0.2): 56 data bytes
64 bytes from 172.21.0.2: seq=0 ttl=64 time=0.109 ms
64 bytes from 172.21.0.2: seq=1 ttl=64 time=0.122 ms

--- A1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.109/0.115/0.122 ms
/ # ping -c 2 A3
PING A3 (172.21.0.4): 56 data bytes
64 bytes from 172.21.0.4: seq=0 ttl=64 time=0.126 ms
64 bytes from 172.21.0.4: seq=1 ttl=64 time=0.077 ms

--- A3 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.077/0.101/0.126 ms

```

```

→ LR2 git:(IGI-LR2) x docker network create -d overlay my-net-overlay
Error response from daemon: This node is not a swarm manager. Use "docker swarm init" or "docker swarm join" to connect this node to swarm and try again.
→ LR2 git:(IGI-LR2) x docker swarm init
Swarm initialized: current node (iqt27svyzftlkhuwz910a5yek) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-5p1as0h4ah0leu26n9uhd3s82j8pzs18dbw8e9qnktrmyjcnwe-b5fviguqx1tx94vpvypscor4x 192.168.0.109:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.

→ LR2 git:(IGI-LR2) x docker network create -d overlay my-net-overlay
4bcu3qyy86ksp1dip7if23te1
→ LR2 git:(IGI-LR2) x docker network ls

```

NETWORK ID	NAME	DRIVER	SCOPE
2dd2caad52b5	bridge	bridge	local
7a8c27f5a335	docker_gwbridge	bridge	local
0830714d6018	host	host	local
n2zeph8rrvny	ingress	overlay	swarm
b3ac90e62daf	mean-stack-registration-login-example_default	bridge	local
9f2cbfd0cc8f	mean-stack-registration-login-example_mean-network	bridge	local
77241a4ca589	my-net-bridge	bridge	local
4bcu3qyy86ks	my-net-overlay	overlay	swarm
24511b00c9e5	my-net-ping	bridge	local
bbd18a792dbf	none	null	local

```

→ LR2 git:(IGI-LR2) x docker network inspect my-net-overlay
[
  {
    "Name": "my-net-overlay",
    "Id": "4bcu3qyy86ksp1dip7if23te1",
    "Created": "2026-02-15T22:06:04.617799399Z",
    "Scope": "swarm",
    "Driver": "overlay",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "10.0.1.0/24",
          "Gateway": "10.0.1.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": null,
    "Options": {
      "com.docker.network.driver.overlay.vxlanid_list": "4097"
    },
    "Labels": null
  }
]

```

```

→ LR2 git:(IGI-LR2) x docker network create -d overlay my-net-overlay2
zi988eac198vswxkd3nb3745d
→ LR2 git:(IGI-LR2) x docker network ls

```

NETWORK ID	NAME	DRIVER	SCOPE
2dd2caad52b5	bridge	bridge	local
7a8c27f5a335	docker_gwbridge	bridge	local
0830714d6018	host	host	local
n2zeph8rrvny	ingress	overlay	swarm
b3ac90e62daf	mean-stack-registration-login-example_default	bridge	local
9f2cbfd0cc8f	mean-stack-registration-login-example_mean-network	bridge	local
77241a4ca589	my-net-bridge	bridge	local
4bcu3qyy86ks	my-net-overlay	overlay	swarm
zi988eac198v	my-net-overlay2	overlay	swarm
24511b00c9e5	my-net-ping	bridge	local
bbd18a792dbf	none	null	local

```

→ LR2 git:(IGI-LR2) x docker network inspect my-net-overlay2
[
  {
    "Name": "my-net-overlay2",
    "Id": "zi988eac198vswxkd3nb3745d",
    "Created": "2026-02-15T22:09:30.433685266Z",
    "Scope": "swarm",
    "Driver": "overlay",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "10.0.2.0/24",
          "Gateway": "10.0.2.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": null,
    "Options": {
      "com.docker.network.driver.overlay.vxlanid_list": "4098"
    },
    "Labels": null
  }
]
→ LR2 git:(IGI-LR2) x docker network rm my-net-overlay2
my-net-overlay2

```

```
→ LR2 git:(IGI-LR2) ✗ docker network create -d host my-net-host
Error response from daemon: only one instance of "host" network is allowed
→ LR2 git:(IGI-LR2) ✗
```

```
→ LR2 git:(IGI-LR2) ✗ docker network prune
WARNING! This will remove all custom networks not used by at least one container.
Are you sure you want to continue? [y/N] y
Deleted Networks:
mean-stack-registration-login-example_mean-network
my-net-ping
docker_gwbridge
mean-stack-registration-login-example_default
my-net-bridge

→ LR2 git:(IGI-LR2) ✗
```